

Chapter 5.2

Marking and labelling of packages including IBCs

Note: These provisions relate essentially to the marking and labelling of dangerous goods according to their properties. However, additional marks or symbols indicating precautions to be taken in handling or storing a package (such as a symbol representing an umbrella, indicating that a package shall be kept dry) may be displayed on a package if appropriate.

5.2.1 Marking of packages including IBCs

5.2.1.1 Unless provided otherwise in this Code, the proper shipping name for the dangerous goods as determined in accordance with 3.1.2 and the corresponding UN number, preceded by the letters "UN", shall be displayed on each package. The UN number and the letters "UN" shall be at least 12 mm high, except for packages of 30 L capacity or less or of 30 kg maximum net mass or less and for cylinders of 60 L water capacity or less when they shall be at least 6 mm in height and except for packages of 5 L capacity or less or of 5 kg maximum net mass when they shall be of an appropriate size. In the case of unpackaged articles, the mark shall be displayed on the article, on its cradle or on its handling, storage or launching device. For goods of division 1.4, compatibility group S, the division and compatibility group letter shall also be marked unless the label for 1.4S is displayed. A typical package mark is:

CORROSIVE LIQUID, ACIDIC, ORGANIC, N.O.S. (caprylyl chloride) UN 3265.

Note: Cylinders of 60 L water capacity or less marked with a UN number in accordance with the provisions of the IMDG Code up to 31 December 2013 and which do not conform to the provisions of 5.2.1.1 regarding the size of the UN number and of the letters "UN" applicable as from 1 January 2014 may continue to be used until the next periodic inspection but no later than 1 July 2018.

5.2.1.2 All package marks required by 5.2.1.1:

- .1 shall be readily visible and legible;
- .2 shall be such that this information will still be identifiable on packages surviving at least three months' immersion in the sea. In considering suitable marking methods, account shall be taken of the durability of the packaging materials used and the surface of the package;
- .3 shall be displayed on a background of contrasting colour on the external surface of the package; and
- .4 shall not be located with other package marks that could substantially reduce their effectiveness.

5.2.1.3 Salvage packagings including large salvage packagings and salvage pressure receptacles shall additionally be marked with the word "SALVAGE". The lettering of the "SALVAGE" mark shall be at least 12 mm high.

5.2.1.4 Intermediate bulk containers of more than 450 L capacity and large packagings shall be marked on two opposing sides.

5.2.1.5 Special marking provisions for radioactive material

5.2.1.5.1 Each package shall be legibly and durably marked on the outside of the packaging with an identification of either the consignor or consignee, or both. Each overpack shall be legibly and durably marked on the outside of the overpack with an identification of either the consignor or consignee, or both unless these marks of all packages within the overpack are clearly visible.

5.2.1.5.2 The marking of excepted packages of radioactive material of class 7 shall be as required by 5.1.5.4.1.

5.2.1.5.3 Each package of gross mass exceeding 50 kg shall have its permissible gross mass legibly and durably marked on the outside of the packaging.

5.2.1.5.4 Each package which conforms to:

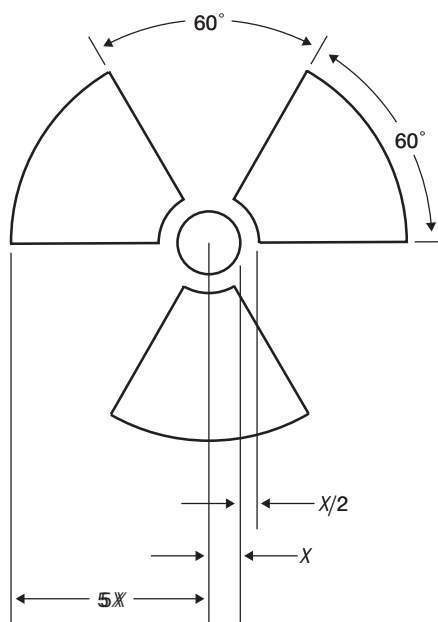
- .1 a Type IP-1 package, a Type IP-2 package or a Type IP-3 package design shall be legibly and durably marked on the outside of the packaging with "TYPE IP-1", "TYPE IP-2" or "TYPE IP-3" as appropriate;

- .2 a Type A package design shall be legibly and durably marked on the outside of the packaging with "TYPE A";
- .3 a Type IP-2 package, a Type IP-3 package or a Type A package design shall be legibly and durably marked on the outside of the packaging with the international vehicle registration code (VRI code) of the country of origin of design and either the name of the manufacturer or other identification of the packaging specified by the competent authority of the country of origin of design.

5.2.1.5.5 Each package which conforms to a design approved under one or more of paragraphs 5.1.5.2.1, 6.4.22.1 to 6.4.22.4, 6.4.23.4 to 6.4.23.7 and 6.4.24.2 shall be legibly and durably marked on the outside of the package with the following information:

- .1 the identification mark allocated to that design by the competent authority;
- .2 a serial number to uniquely identify each packaging which conforms to that design;
- .3 "Type B(U)", "Type B(M)" or "Type C", in the case of a Type B(U), Type B(M) or Type C package design.

5.2.1.5.6 Each package which conforms to a Type B(U), Type B(M) or Type C package design shall have the outside of the outermost receptacle which is resistant to the effects of fire and water plainly marked by embossing, stamping or other means resistant to the effects of fire and water with the trefoil symbol shown below.



Basic trefoil symbol with proportions based on a central circle of radius X.
The minimum allowable size of X shall be 4 mm.

Any mark on the package made in accordance with the requirements of 5.2.1.5.4.1 and .2 and 5.2.1.5.5.3 relating to the package type that does not relate to the UN number and proper shipping name assigned to the consignment shall be removed or covered.

5.2.1.5.7 Where LSA-I or SCO-I material is contained in receptacles or wrapping materials and is transported under exclusive use as permitted by 4.1.9.2.4, the outer surface of these receptacles or wrapping materials may bear the mark "RADIOACTIVE LSA-I" or "RADIOACTIVE SCO-I", as appropriate.

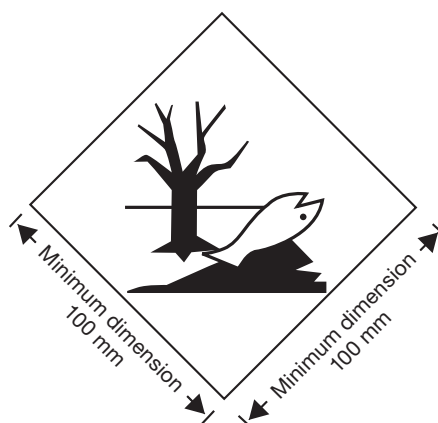
5.2.1.5.8 In all cases of international transport of packages requiring competent authority approval of design or shipment, for which different approval types apply in the different countries concerned by the shipment, marking shall be in accordance with the certificate of the country of origin of the design.

5.2.1.6 Special marking provisions for marine pollutants

5.2.1.6.1 Except as provided in 2.10.2.7, packages containing marine pollutants meeting the criteria of 2.9.3 shall be durably marked with the marine pollutant mark.

5.2.1.6.2 The marine pollutant mark shall be located adjacent to the marks required by 5.2.1.1. The provisions of 5.2.1.2 and 5.2.1.4 shall be met.

5.2.1.6.3 The marine pollutant mark shall be as shown in the figure below.



Marine pollutant mark

The mark shall be in the form of a square set at an angle of 45° (diamond-shaped). The symbol (fish and tree) shall be black on white or a suitable contrasting background. The minimum dimensions shall be 100 mm × 100 mm and the minimum width of line forming the diamond shall be 2 mm. If the size of the package so requires, the dimensions/line thickness may be reduced, provided the mark remains clearly visible. Where dimensions are not specified, all features shall be in approximate proportion to those shown.

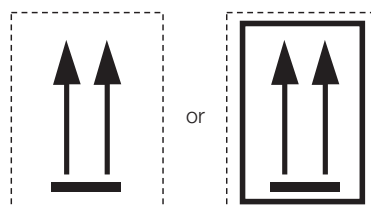
Note: The labelling provisions of 5.2.2 apply in addition to any requirement for packages to bear the marine pollutant mark.

5.2.1.7 Orientation arrows

5.2.1.7.1 Except as provided in 5.2.1.7.2:

- combination packagings having inner packagings containing liquid dangerous goods;
- single packagings fitted with vents;
- closed or open cryogenic receptacles intended for the transport of refrigerated liquefied gases; and
- machinery or apparatus containing liquid dangerous goods when it is required to ensure the liquid dangerous goods remain in their intended orientation (see special provision 301 of chapter 3.3),

shall be legibly marked with package orientation arrows which are similar to the illustration shown below or with those meeting the specifications of ISO 780:1997. The orientation arrows shall appear on two opposite vertical sides of the package with the arrows pointing in the correct upright direction. They shall be rectangular and of a size that is clearly visible commensurate with the size of the package. Depicting a rectangular border around the arrows is optional.



Two black or red arrows on white or suitable contrasting background.

The rectangular border is optional.

All features shall be in approximate proportion to those shown.

5.2.1.7.2 Orientation arrows are not required on:

- .1 outer packagings containing pressure receptacles except closed or open cryogenic receptacles;
- .2 outer packagings containing dangerous goods in inner packagings each containing not more than 120 mL, with sufficient absorbent material between the inner and outer packagings to completely absorb the liquid contents;
- .3 outer packagings containing Division 6.2 infectious substances in primary receptacles each containing not more than 50 mL;
- .4 type IP-2, type IP-3, type A, type B(U), type B(M) or type C packages containing class 7 radioactive material;
- .5 outer packagings containing articles which are leak-tight in all orientations (e.g. alcohol or mercury in thermometers, aerosols, etc.); or

- .6 outer packagings containing dangerous goods in hermetically sealed inner packagings each containing not more than 500 mL.

5.2.1.7.3 Arrows for purposes other than indicating proper package orientation shall not be displayed on a package marked in accordance with this subsection.

5.2.1.8 **Excepted quantity mark**

5.2.1.8.1 Packages containing excepted quantities of dangerous goods shall be marked according to 3.5.4.

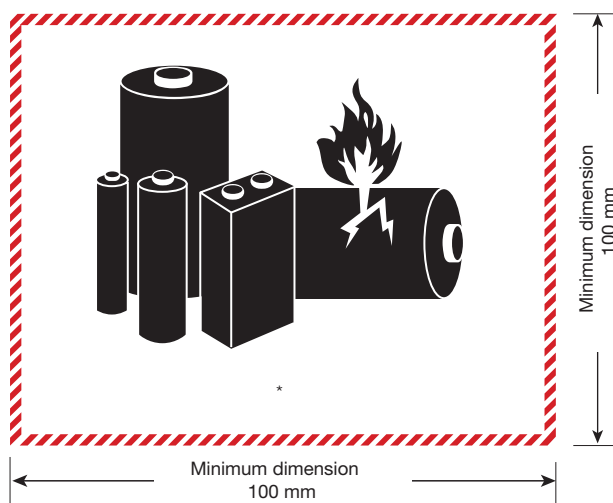
5.2.1.9 **Limited quantity mark**

5.2.1.9.1 Packages containing dangerous goods packed in limited quantities shall be marked according to 3.4.5.

5.2.1.10 **Lithium or sodium ion battery mark**

5.2.1.10.1 Packages containing lithium or sodium ion cells or batteries prepared in accordance with special provision 188 shall be marked as shown in figure below.

5.2.1.10.2 The mark shall indicate the UN number, preceded by the letters "UN", i.e. 'UN 3090' for lithium metal cells or batteries, 'UN 3480' for lithium ion cells or batteries, or 'UN 3551' for sodium ion cells or batteries. Where the cells or batteries are contained in, or packed with, equipment, the UN number preceded by the letters "UN", i.e. 'UN 3091', 'UN 3481' or 'UN 3552' as appropriate shall be indicated. Where a package contains cells or batteries assigned to different UN numbers, all applicable UN numbers shall be indicated on one or more marks.



△ **Lithium or sodium ion battery mark**

* Place for UN number(s)

The mark shall be in the form of a rectangle or a square with hatched edging. The dimensions shall be a minimum of 100 mm wide × 100 mm high and the minimum width of the hatching shall be 5 mm. The symbol (group of batteries, one damaged and emitting flame, above the UN number(s)) shall be black on white or suitable contrasting background. The hatching shall be red. If the size of the package so requires, the dimensions may be reduced to not less than 100 mm wide × 70 mm high. Where dimensions are not specified, all features shall be in approximate proportion to those shown.

Note: The mark shown in the figure "Lithium battery mark" in 5.2.1.10.2 of the IMDG Code Amendment 40-20, showing the telephone number for additional information, may continue to be applied until 31 December 2026.

5.2.2 **Labelling of packages including IBCs**

5.2.2.1 **Labelling provisions**

These provisions are related essentially to danger labels. However, additional marks or symbols indicating precautions to be taken in handling or storing a package (such as a symbol representing an umbrella, indicating that a package shall be kept dry) may be displayed on a package if appropriate.